

DARE Multi-Agent Workflows for Research Synthesis and Meta- Analysis (SALSA)

Dahirou Dabo Harden
Georgia Tech Open Source Program Office

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About DARE

- Developed by the Carnegie Mellon University Open Source Program Office
- Dietrich Analysis Research Education (DARE) is a comprehensive platform that functions as a "large language model (LLM) server starter pack" for higher education. DARE provides direct API access to LLMs while maintaining an accessible interface that supports progressive skill development and emphasizes transparency and user agency in human-AI collaboration.
- With DARE, you have access to,
 - Anthropic Claude Models, OpenAI models and other open-source alternatives
 - This now includes GT meta-llama/Llama-3.1-8B-Instruct resulting from our joint efforts with DARE team
 - Secure file storage with advanced Retrieval Augmented Generation (RAG) capabilities
 - Customizable system prompts for consistent AI interactions
 - Cost tracking with detailed token usage monitoring and billing transparency

Read more
at: <https://www.cmu.edu/dietrich/ai/education/dare.html>



GT SPIN/Collaboration with DARE

Developed by the GT PACE team

SPIN stands for: The Service for Persistent Inference

SPIN provides on-prem inference models with API endpoints

- Backend composed of LiteLLM and vLLM
- Models run on Hopper GPUs on PACE's instructional cluster
- Models include Llama 3.1, Qwen2.5 VL, and Qwen3 Embeddings, with plans to expand offerings
- Provides an OpenAI-compatible API endpoint, which is what DARE uses

Architectural View of SPIN (diagram on right)

LiteLLM

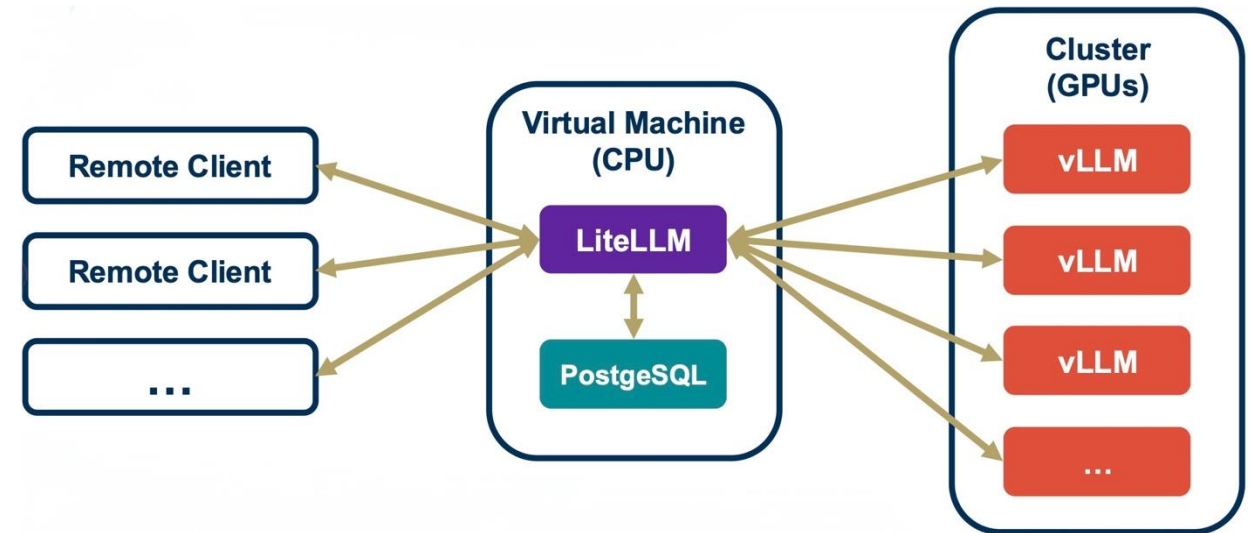
- Reverse proxy between client and GPU resources
- Handles per-user and per-team key provisioning, usage accounting, rate limits, ...
- Backed by a PostgreSQL database for accounting and usage

vLLM

- Actually does the inference on GPUs
- One vLLM instance serves one model on 1+ GPU
- Instances are "persistent" (continuously scheduled by Slurm)

Clients

- All type of clients interact via API requests with key-based authentication



*This diagram is of an earlier prototype

Clicking "enter" into the conversation box on DARE...

1. DARE acts as the remote client from SPIN's perspective
2. Request goes to LiteLLM
3. LiteLLM authenticates the key, checks model mapping/accounting
4. LiteLLM forwards to the correct vLLM instance
5. vLLM runs inference on the GPU node
6. Output returns through LiteLLM back to DARE

What is SALSA?

SALSA was introduced by M. J. Grant and A. Booth in 2009 as a simple framework for describing review methods using four steps:

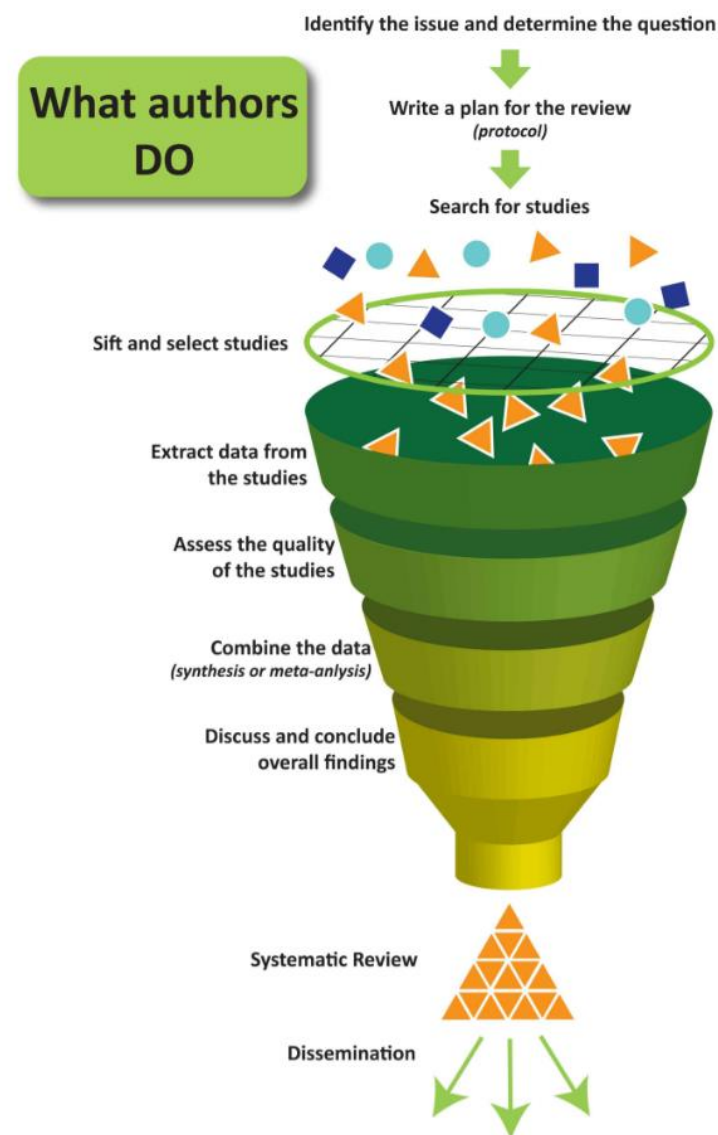
- **Search:** identify and collect relevant studies
- **Appraisal:** evaluate quality and relevance of those studies
- **Synthesis:** combine extracted findings across studies
- **Analysis:** interpret results, patterns, and implications

In today's demo, we use synthesis rather than meta-analysis, since meta-analysis requires directly comparable quantitative results from sufficiently similar studies and evaluations.

Question: "How do existing graph analytics methods compare in terms of performance, scalability, and applicability across domains?"

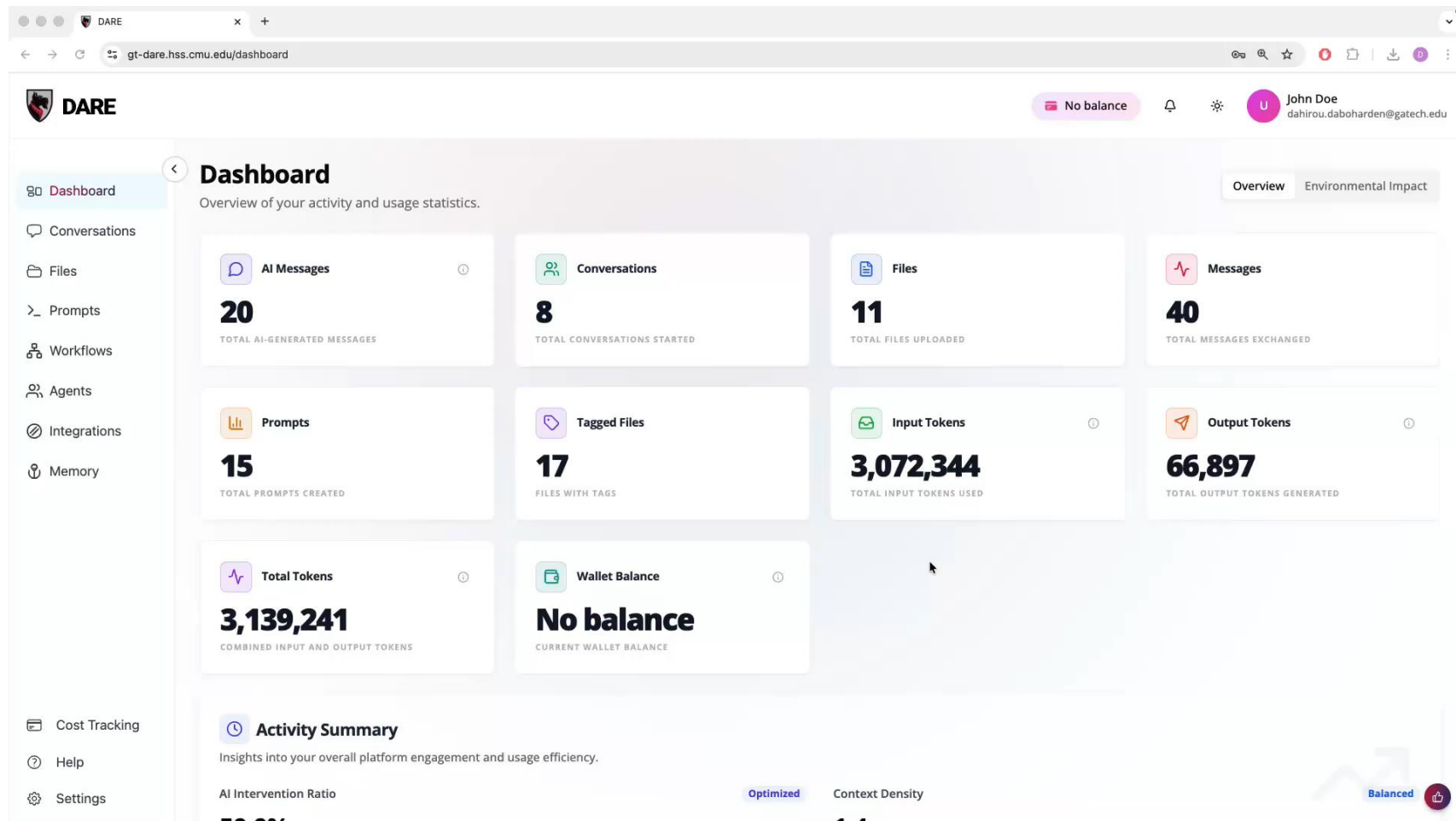
More about the Demo

- All on the general topic of graph analytics, not fixed to a tight field
- Total tokens across all docs: ~51k
- For reference
 - Meta-llama 3.1-8B avg token to word conversion is ~0.75:1
 - Meta-llama 3.1-8B context window: 128k
- What is compared among our small pool:
 - Task
 - Datasets
 - Methods/baselines
 - Evaluation setup
 - Outcomes like performance, scalability, and runtime



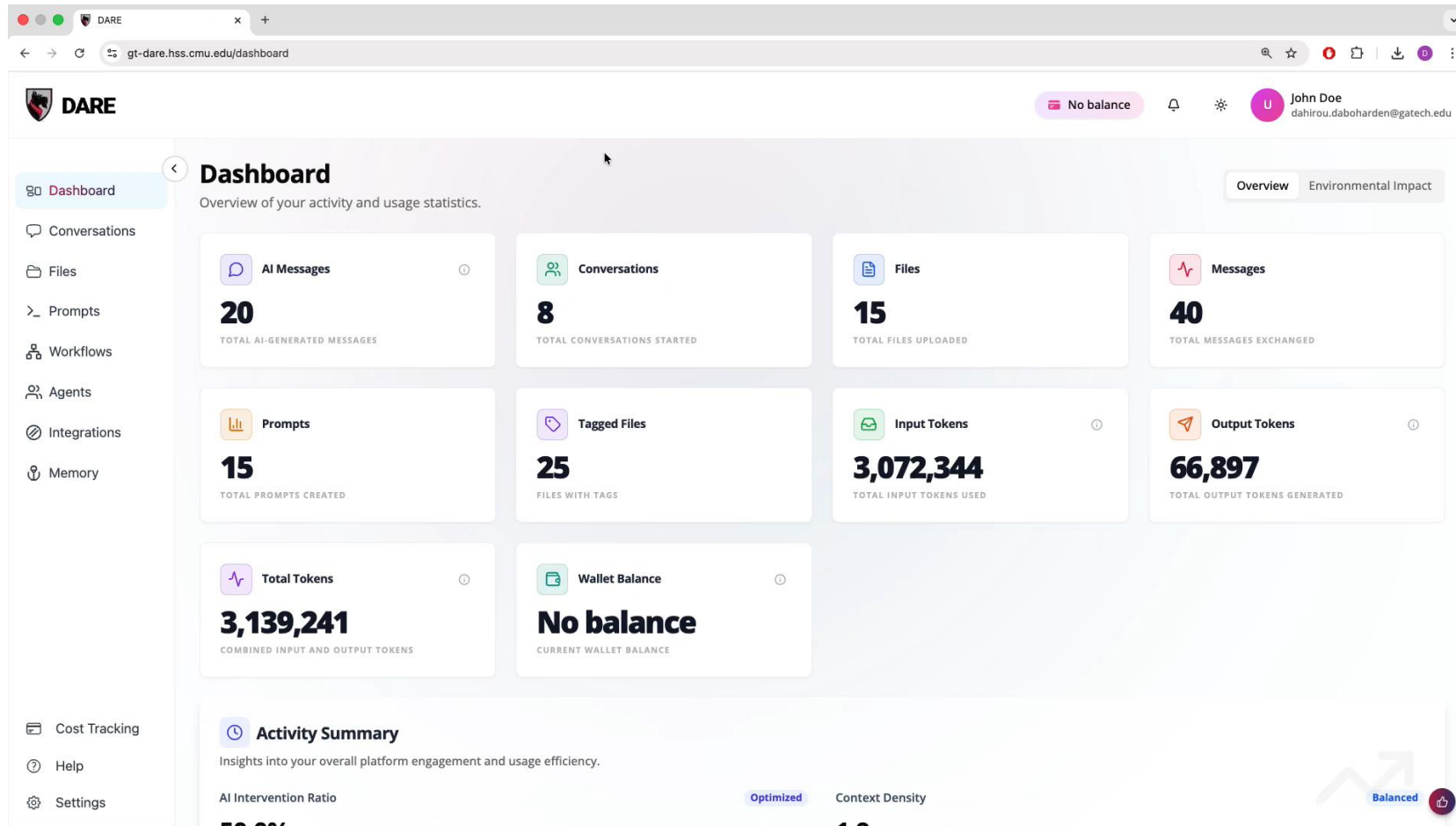
Demo

1. Upload files to DARE



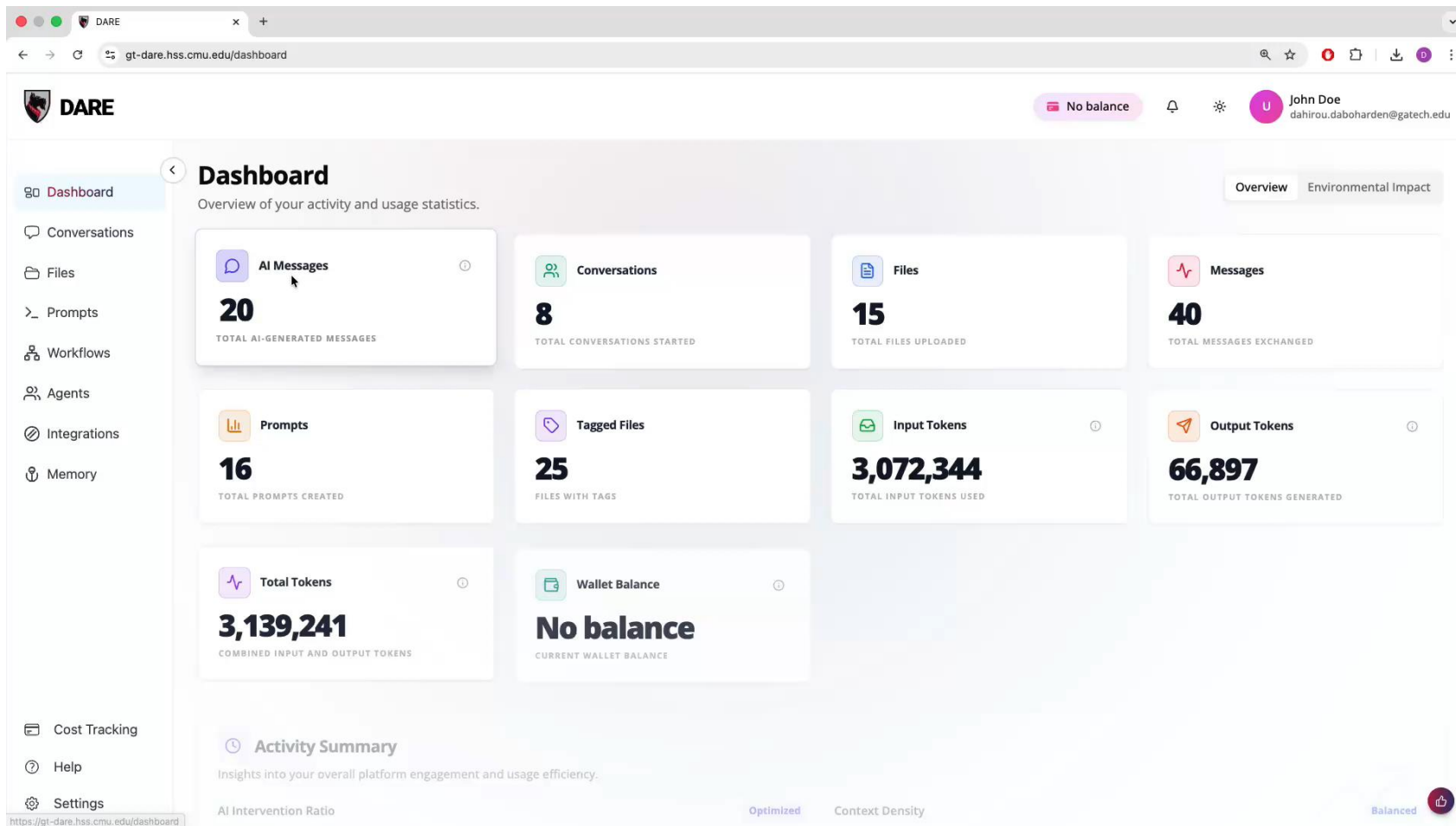
Demo

2. Create SALSA prompt



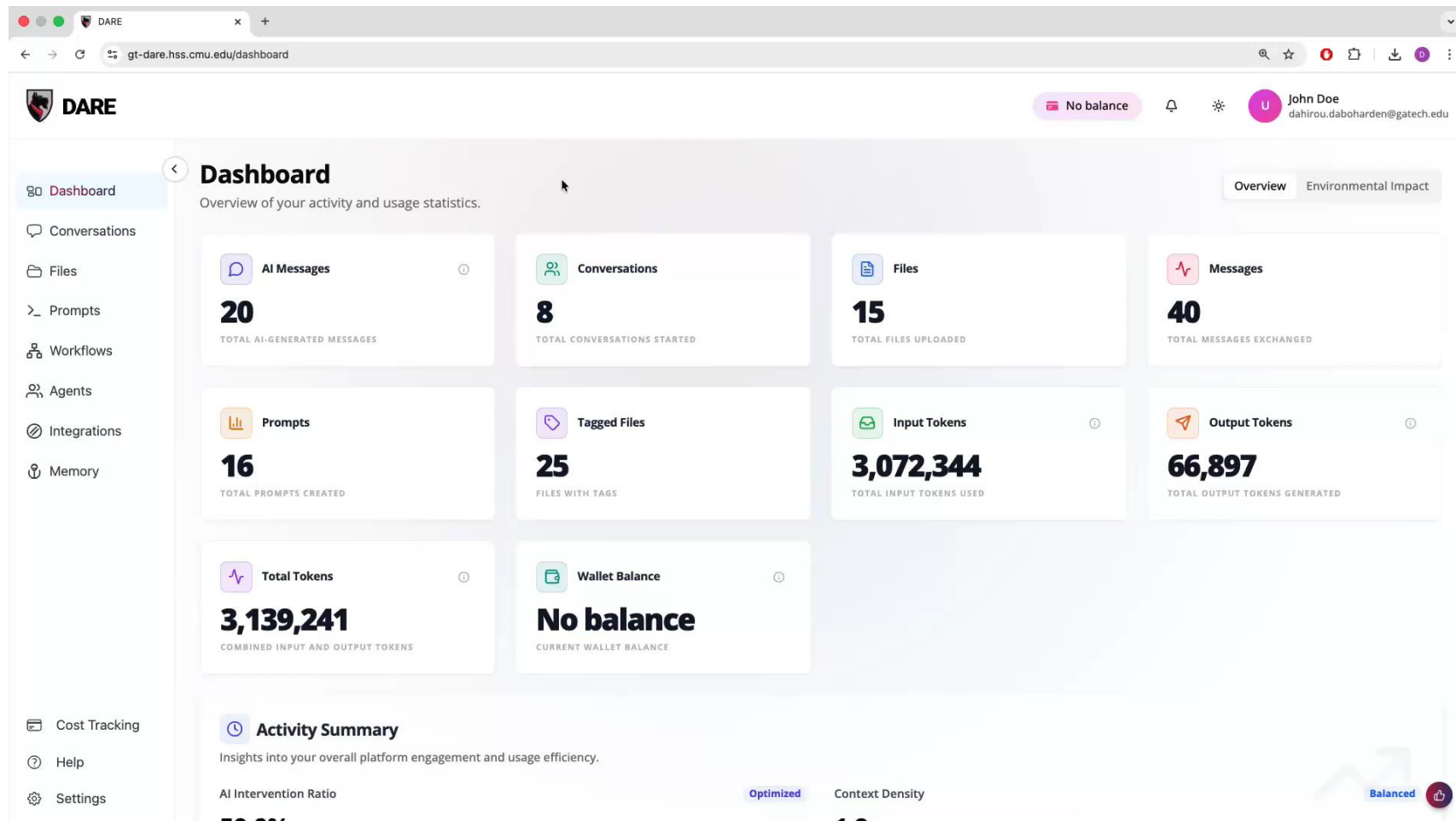
Demo

3. Create agents



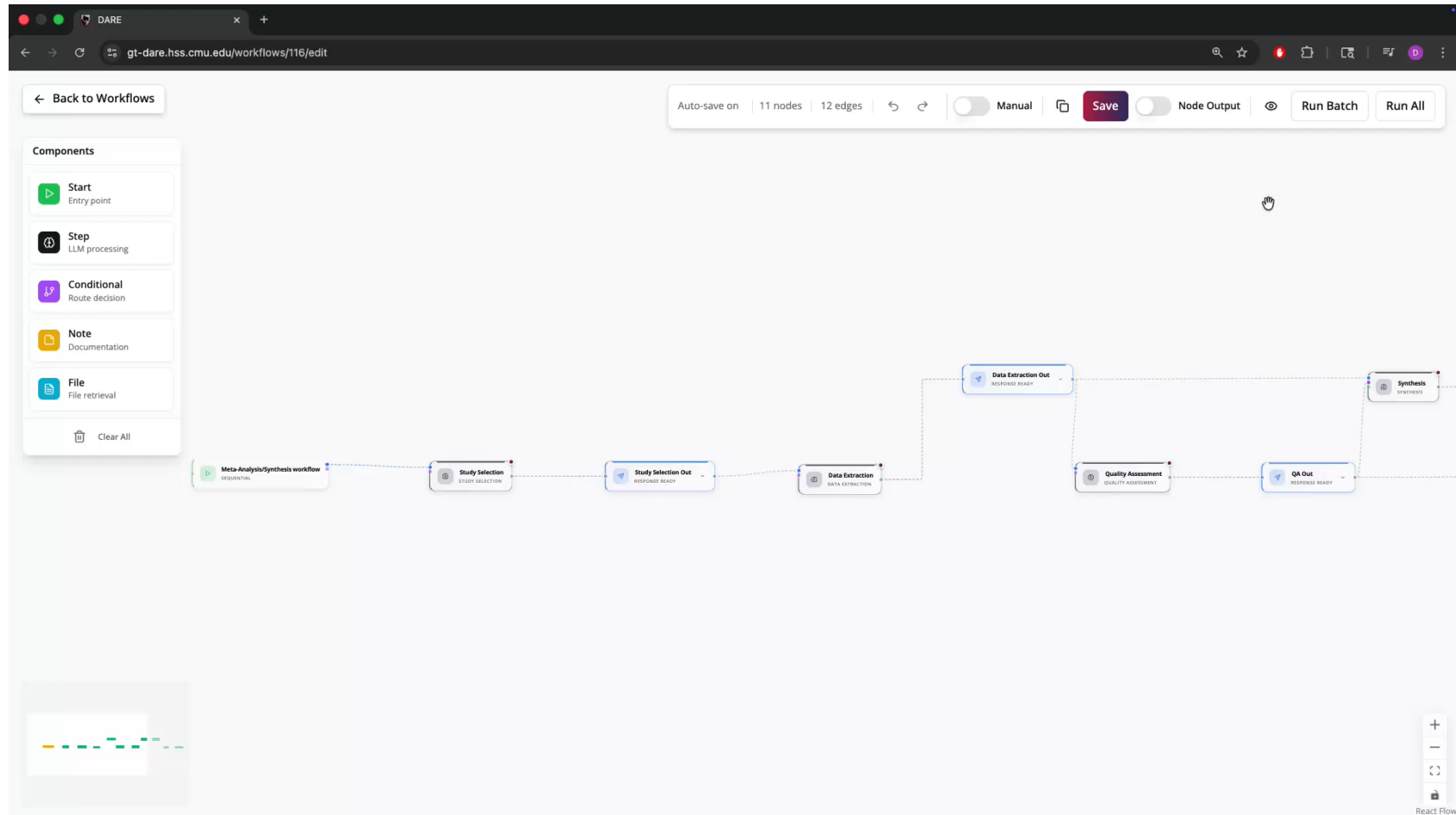
Demo

4. Create workflow



Demo

5. Run workflow



Thanks for your attention!

Questions?

References

1. <https://pace.gatech.edu/>
2. **Rahaman, R., Liu, F. etc,** (2026). Service-Oriented Architecture for an On-Premises AI Gateway Without an On-Premises Cloud. To be appeared In Practice and Experience in Advanced Research Computing 2026
3. Grant, M. J., & Booth, A. (2009). A typology of reviews: An analysis of 14 review types and associated methodologies. Health Information & Libraries Journal, 26(2), 91–108. <https://doi.org/10.1111/j.1471-1842.2009.00848.x>
4. Ihdego, B., Ahmed, M., Dermody, K., Farnum, C., Jakubek, D., Petropoulos, J.-A., Schmidt, J., & Steinberg, R. (2022, February 28). *Conducting a systematic review*. Advanced Research Skills Conducting Literature and Systematic Reviews. <https://pressbooks.library.torontomu.ca/graduaterreviews/chapter/conducting-a-systematic-review/>